



The Master Thesis with Title

FAULT PREDICTION IN SEMICONDUCTOR MANUFACTURING USING XGBOOST AND RANDOM FOREST MODELS

Submitted to

Information Technology Department, Sana'a University

In Partial Requirements of

Master's Degree

In

Information Technology

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2026 / 1447

Abstract

Fault detection is a challenge in industrial environments due to its complexity and sensitivity. This research presents an advanced methodology using machine learning (ML) to predict faults in the semiconductor manufacturing industry. The study aims to reach the highest accuracy in predicting faults during manufacturing. It used datasets called UCI-SECOM, which were taken from Kaggle. The dataset consists of missing values and unbalanced classes.

The study introduced a holistic methodology that has an accurate processing for the data and deals with the problem of unbalanced classes, applying strong machine learning (ensemble models). The proposed methodology, compared to other researches, achieved high accuracy with zero false negatives for the faulty class. The main factors for achieving this high accuracy is compensating for missing value using the median imputation, eliminating the unnecessary pattern, using the SMOTE technique to increase the artificial samples, processing the unbalanced samples, and using enhanced XGBoost and Random Forest models. The ensemble model was evaluated using various performance metrics, showing an AUC of 99.48%. Predicting faults in the semiconductor manufacturing industry in the early stages can save time and resources.

Keywords: UCI-SECOM, machine learning (ML), semiconductor manufacturing, XGBoost, Random Forest.

Contents

Abstract	i
1 Introduction	1
1.1 Problem Statement	2
1.2 Purpose Statement	2
1.3 Research Questions	2
1.4 Research Objectives	3
1.5 Significance of the Study	3
1.6 Study Limitations	3
1.7 Definition of Key Terms	4
1.8 Brief Methodology Overview	5
1.9 Organization of the Thesis	5
2 Literature Review	7
2.1 Introduction	7
2.2 Semiconductor Fault Prediction Approaches	7
2.3 Machine Learning Techniques	7
2.4 Studies Utilizing the SECOM Dataset	8
2.4.1 Comparative Analysis of Selected Studies on SECOM Dataset	8
2.5 Conclusion of Literature Review	9
2.6 Research Gap	9
3 Methodology	11
3.1 Introduction	11
3.2 Study Nature and Methodology Family Identification	11
3.3 Research Design and Execution Pathway	11
3.3.1 Design Type	11
3.3.2 Unit of Analysis	11
3.3.3 Execution Pathway & Environment	11
3.3.4 Rationale for Design Choice	12
3.4 Data and Measurement Procedures	12
3.4.1 Dataset Description	12
3.4.2 Data Preprocessing and Cleaning	12
3.4.3 Model Training	12
3.4.4 Ensemble Learning	12
3.4.5 Performance Evaluation	13
3.5 Expected Outcomes	13
4 References	14

List of Tables

1	COMPARISON BETWEEN METHODOLOGIES	8
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List of Figures

Chapter 1

INTRODUCTION

1 Introduction

In recent years, the industrial revolution seen has changed completely in the industrial environments, the huge industrial data, and the smart analytics becomes a very basic section to improve the productivity and reduce the predictive maintenance considered as one of the famous or most important applications to this transformation. It aims to predict the maintenance earlier, before it happens, through the historical data analytics and sensitive data. This contribution could reduce the unplanned downtime, to increase the productivity, and enhances the industrial products quality [8], [10]. Semiconductor manufacturing is one of the most complex manufacturing environments due to the depending on very sensitive processes, multi-stages processes, and hundreds of sensors which generate a high dimensional data. This complexity makes the fault prediction maintenance huge challenge, especially with the unbalanced data. The fault cases are too small in compared with the normal one [2], [11]. Several studies clarify the importance of Machine Learning techniques in the smart manufacturing processing data. It mentioned the problem of Rare Class Prediction in the smart industrial environments, highlighting the importance of improving detection performance for critical yet infrequent events [1]. other studies introduced an experimental evaluation to diagnoses the fault in un balanced and completed data it demonstrated that the performance effected strongly by missing values and skewed class distributions [2]. Other studies focus on building a predictive fault detection model using different algorithm It devolved a predictive model for semiconductor manufacturing processes [4]. while other study, depend on Machine Learning techniques to build predictive system for semiconductor failures [12]. some research discussed the impact of unbalanced datasets on predictive maintenance in semiconductor production environments [11]. studies presented a comprehensive survey on predictive maintenance in Industry 4.0 clarifying the challenges of data quality, unbalance, and scalability as major research challenges [8]. On the other hand, studies confirmed this results remaining the necessity of developing advanced new strategies to enhancing the unbalanced data [10]. during the revolution in deep learning techniques a new developed solution appears based on meta learning and purning models to detect the in industrial processes [6]. Other researches ensures that the data Preprocessing can enhanced the performance of of ML models in semiconductor manufacturing specially with missing values and addressing high dimensionality [14]. Additional contributions in manufacturing systems shows predictive maintenance concept with a holistic view [3]. In addition, other researches ensures that integration of the smart analytical into industrial environments can reduces the financial loses and improving the continues of the business [5]. Studies emphasized the important of integration of predictive maintenance with production scheduling to optimize manufacturing performance [13]. Although the improvement in predictive maintenance reached a high level, but several challenges still faced researches such as the accurate prediction of rare failure events in highly imbalanced and high-dimensional semiconductor datasets continues to limit the reliability of existing models. Performance metrics such as Recall and F1-score often decline when detecting minority

fault classes, indicating the need for more robust and balanced predictive frameworks [1], [2], [11], [14]. Therefore, this study aims to develop a predicted model based on advanced machine learning capable of effectively handling imbalanced data, improving rare fault detection. That can enhance the predictive performance in semiconductor manufacturing environments which will save manufacturing industrial's money and resources. So this chapter summaries the problem of the fault detection in manufacturing environments.

1.1 Problem Statement

Despite The continuous increasing in using machine learning techniques in semiconductor industry and although the continuous development in the fourth industry, however, the rare fault predictive in manufacturing in multi-stages manufacturing still represented as a huge challenge. Studies contribute to develop the predicting accuracy and increasing the performance indicators and its metrics, and that is so hard especially with the sensitivity of the semiconductor manufacturing data. Because of its high-dimensional and unbalanced characteristics. tend to be biased toward the majority class, achieving high overall accuracy while failing to detect critical rare faults. In addition, traditional faulty detected methods diagnose the faults after the manufacturing stages[15]. That increases operational costs and delays corrective actions. The recent predicted models face difficulties in balancing between the detective performance and the capability dealing with the rare samples in the complex manufacturing environments. For that, this study presents the needs of developing a predictive model can handle the old previous problems and can improve the predicted quality, which will decrease resource cost and saving the time in semiconductor manufacturing systems.

1.2 Purpose Statement

The purpose of this quantitative study is to predict equipment failures in semiconductor manufacturing systems using XGboost, Random Forest models and the SECOM dataset with the aim of improving prediction accuracy.

1.3 Research Questions

1. How can machine learning improve the accuracy used for predicting equipment failures in semiconductor manufacturing systems?
2. How can the optimization strategies, besides the models, improve the performance matrix?
3. What are the key factors that contribute to exceed the accuracy (99.00%) in semiconductor fault prediction?

1.4 Research Objectives

- To exceed 99% predictive accuracy.
- To examine the ensemble models for optimizing the failures prediction.
- To examine the efficiency of the SMOTE and cleaning missing value.

1.5 Significance of the Study

This study is significant at several sides and for several kind of people:

A. For semiconductor manufacturing: Semiconductor is a very sensitive data it effected with any external factor This study introduced a methodology that capable to exceed failure predicting 99%, and introduced a path for reducing manufacturing costs and improving the quality of products and its reliability. This can support the productivity in general, and it can early detect overall before crossing all the stages which will prevent wasting resources and saving time and money. In addition, that can improve client satisfaction.

B. For machine learning researchers: This study contributes on supporting the knowledge of applying advance machine learning techniques using real manufacturing datasets, which is complicated and unbalanced. It focuses on an efficient strategy that process the data and dealing with the unbalanced classes. These strategies can be generalized to other or similar problems in other domains.

C. For Data Scientists and Engineers The detailed methodology introduced a practical example and steps of cleaning data techniques features engineering, choosing the models, and hypermeter tuning in the high-performance predicted maintenance systems. It contains the steps that are use and the results that can be reused and implemented.

D. For Academic Research: This research provides a benchmark for fault prediction accuracy on used SECOM dataset, encouraging further exploration and development of even more robust and interpretable models

1.6 Study Limitations

The study focuses on fault prediction in semiconductor manufacturing using SECOM dataset which downloaded from Kaggle. The scope includes:

- **Data Source:**
the SECOM dataset contains 1567 samples and 591 numerical features.

- **Methodology:**

Python and advanced machine learning techniques, including data preprocessing (missing value imputation, feature selection), class imbalance handling (SMOTE), and supervised learning models :XGBoost, Random Forest.

- **Evaluation:**

Performance evaluation primarily based on accuracy, precision, recall, F1-score, and confusion matrix.

1.7 Definition of Key Terms

A. Semiconductor: A material with electrical conductivity between that of a conductor and an insulator, used in electronic components.

B. Fault Prediction: The process of identifying potential defects before they occur.

C. Machine Learning (ML): A field of artificial intelligence that enables systems to learn from data without being explicitly programmed.

D. Deep Learning (DL): A subset of machine learning that uses artificial neural networks with multiple layers to learn complex patterns.

E. SECOM Dataset TA publicly available dataset from the UCI Machine Learning Repository and Kaggle, containing sensor readings and process parameters from a semiconductor manufacturing process, used for fault detection.

F. Class Imbalance: A situation in classification problems where the number of observations belonging to one class is significantly lower than those belonging to other classes.

G.SMOTE (Synthetic Minority Over-sampling Technique): An oversampling technique used to address class imbalance by generating synthetic samples for the minority class.

H. XGBoost (Extreme Gradient Boosting): An optimized distributed gradient boosting library designed to be highly efficient, flexible, and portable.

I. Random Forest: An ensemble learning method for classification and regression that operates by constructing a multitude of decision trees at training time and outputting the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees.

J. Accuracy: The proportion of correct predictions among the total number of cases examined.

K. Precision: The ratio of correctly predicted positive observations to the total predicted positive observations.

L. Recall: The ratio of correctly predicted positive observations to all observations in the actual class.

M. F1-Score: The weighted average of Precision and Recall, useful for imbalanced datasets.

N. Confusion Matrix: A table used to describe the performance of a classification model on a set of test data for which the true values are known

1.8 Brief Methodology Overview

The methodology that used in this study conclude several steps to achieve a high fault prediction accuracy. It used many techniques to reach to this accuracy, contains using sacom dataset and handling missing values through median imputation and reducing dimensionally by removing features with low variance. And the issue of class imbalance is addressed using SMOTE technique to create more balanced dataset for training. The study used two machine learning models, XGBoost and Random Forest, which they are trained and fine-tuned using the processed and balanced data. Their performance is evaluated using a suite of metrics, such as accuracy, precision, recall, F1 score. In addition, the confusion matrix with a specific focus on the minority class detection.

1.9 Organization of the Thesis

This thesis is organized into three main chapters:

Chapter 1: Introduction This chapter provides an overview of the research problem, the purpose and significance of the study, outlines the research questions and objectives, defines the scope and limitations, introduces key terminology, and presents a brief overview of the methodology and the organization of the thesis.

Chapter 2: Literature Review: This chapter presents a comprehensive review of existing literature on semiconductor fault prediction, machine learning techniques applied in this domain, and studies utilizing the SECOM dataset. It will critically analyze previous research, identify gaps, and compare reported accuracies, including studies that achieved around 98% accuracy, to contextualize the current study's contribution.

Chapter 3: Methodology : This chapter details the practical implementation of the fault prediction system. It describes the data preprocessing steps, the application of class imbalance handling techniques, the architecture and training of the chosen machine learning models (XGBoost and Random Forest), and the hyperparameter tuning process.

Chapter 2

LITERATURE REVIEW

2 Literature Review

2.1 Introduction

The rapid evolution of the semiconductor industry necessitates robust fault prediction mechanisms to ensure product quality, reduce manufacturing costs, and maintain competitive advantage. This chapter provides a comprehensive review of existing literature on semiconductor fault prediction, with a particular focus on studies employing machine learning techniques and utilizing the SECOM dataset. The review aims to contextualize the current research by identifying prevalent methodologies, reported performance metrics, and areas where further improvements, such as achieving over than 99% accuracy, can make a significant contribution.

2.2 Semiconductor Fault Prediction Approaches

Historically, fault detection in semiconductor manufacturing relied on statistical process control (SPC) and expert systems. While effective for well-understood processes, these methods often struggle with the increasing complexity, high dimensionality, and dynamic nature of modern semiconductor fabrication [1]. The sheer volume of sensor data and process parameters generated during manufacturing makes manual analysis impractical and prone to errors. In recent years, machine learning (ML) and deep learning (DL) have emerged as powerful tools for fault prediction in this domain [2]. These data-driven approaches can automatically learn complex patterns and relationships from historical data, enabling more accurate and proactive identification of potential defects. Studies have shown that ML models can effectively handle the high-dimensional nature of semiconductor data and adapt to subtle shifts in manufacturing processes [3].

2.3 Machine Learning Techniques

A wide array of machine learning algorithms has been applied to semiconductor fault prediction. Common approaches include: Support Vector Machines (SVMs): Known for their effectiveness in high-dimensional spaces and ability to find optimal separating hyperplanes [4]. Decision Trees and Ensemble Methods: Algorithms like Random Forest and Gradient Boosting (e.g., XGBoost) are popular due to their robustness, ability to handle various data types, and capacity to capture non-linear relationships [5] [6]. These methods often provide good interpretability through feature importance scores. Neural Networks and Deep Learning: Artificial Neural Networks (ANNs) and Deep Learning models, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), are increasingly used for their ability to learn hierarchical features from raw data, especially in complex scenarios involving multimodal data [7] [8]. Despite the advancements, challenges persist, primarily related to data preprocessing, feature engineering, and handling class imbalance, which is a pervasive issue in fault detection

where faulty instances are significantly outnumbered by normal ones [9].

2.4 Studies Utilizing the SECOM Dataset

The SECOM dataset, originating from a real-world semiconductor manufacturing process, has become a benchmark for research in fault prediction due to its realistic characteristics, including high dimensionality (591 features), missing values, and severe class imbalance (only 104 faulty instances out of 1567) [10] [11]. Numerous studies have attempted to achieve high accuracy on this dataset, employing various methodologies. For instance, a study by El Mourabit et al. (2020) explored different machine learning approaches on the SECOM dataset, achieving an accuracy of 91% with an MLP predictive model after applying PCA for feature reduction [5]. Another research by Chazhoor and Mounika (2020) focused on predictive maintenance using classification models, reporting accuracies over 98% with Random Forest, highlighting the effectiveness of ensemble methods [9]. This demonstrates that with appropriate preprocessing and model selection, high performance is attainable. More recently, Park and Kim (2024) investigated data preprocessing techniques for machine learning based on SECOM data, emphasizing the importance of imputation and scaling for improving model performance [12]. Cho et al. (2023) achieved remarkable results by addressing the highly imbalanced nature of semiconductor chip-level defects using multimodal fusion and logit adjustment, contributing significantly to the field [1]. Their work, among others, underscores the critical role of advanced techniques in pushing the boundaries of prediction accuracy.

2.4.1 Comparative Analysis of Selected Studies on SECOM Dataset

In this section, Table 1 provides a comparative overview of selected research efforts on the SECOM dataset, highlighting their methodologies and reported performance.

Table 1: COMPARISON BETWEEN METHODOLOGIES

Ref	Methodology	Accuracy	Precision	Recall	F1-score
[1]	Rare Class Prediction Model (ML)	0.95	0.66	0.96	—
[14]	SVM + ADASYN + MaxAbs scaling	85.99%	—	—	—
[11]	(SVC)	98.06%	98.06%	98.06%	98.06%
[12]	(Naive Bayes, C4.5 DT, MLP, SVM)	91% (MLP best)	—	—	—

This comparison demonstrates that many studies achieve high accuracies, but the combination of robust preprocessing, effective class imbalance handling (like SMOTE), and powerful

ensemble models (like Random Forest and XGBoost) plays a essential role in improving the performance of prediction. In addition, the ability of detecting faulty instances, particularly through achieving high recall for the minority class, it's a basic requirement in semiconductor manufacturing environments.

2.5 Conclusion of Literature Review

The literature review highlights the critical need for accurate fault prediction in semiconductor manufacturing and the increasing reliance on machine learning techniques to address this challenge. The SECOM dataset serves as a valuable resource for benchmarking these efforts. While various studies have achieved commendable results, particularly those employing ensemble methods and advanced data preprocessing, there remains a continuous drive to push the boundaries of prediction accuracy and model robustness, especially in handling severe class imbalance. This study builds upon these foundations by demonstrating a methodology that expected to achieves an exceptional 99

2.6 Research Gap

Although the progress achieved in semiconductor fault prediction which used in machine learning techniques, but there are limitations still available in existing literature. Many studies reported high accuracy results, but these results are often influenced by the highly imbalanced nature of the SECOM dataset, where non-faulty instances dominate. Maybe these models achieved high accuracy but at the same time it still performing inadequately in detecting minority (faulty) instances. Furthermore, a considerable number of studies focus primarily on individual components of the prediction pipeline, such as preprocessing techniques, feature selection, or model selection, without integrating these elements to a unified improved framework. Limited attention has been given to balancing multiple performance metrics simultaneously, particularly precision and recall for the minority class, which are critical in real-world semiconductor manufacturing environments. Therefore, there is a clear need for a more comprehensive and robust approach that effectively integrates preprocessing, class imbalance handling, and advanced machine learning models to achieve both high accuracy and reliable fault detection performance.

Chapter 3

METHODOLOGY AND RESULTS

3 Methodology

3.1 Introduction

This section introduces a methodological framework for developing a robust fault detection system using the UCI-SECOM dataset which have been built based on theoretical foundations and literature gaps that mentioned in Chapters 1 and 2. This chapter moved from theoretical information to practical. the methodology consists of steps designed to achieve high classification accuracy in Semiconductor manufacturing. The researcher used different models and assistance tools to reach a high accuracy rate.

3.2 Study Nature and Methodology Family Identification

This study refers to the Analytics Family that based on Data-driven Studies. This research based on analyzing sensor events captured from an industrial environment.

- **Unit of Analysis:** The unit of analysis is a "Single Manufacturing Run" each record represents a specific production instance
- **Data Source Type** The research depends on Secondary Data (Log-based), which refer to machine learning workflows, because we use real Semiconductor manufacturing data.

3.3 Research Design and Execution Pathway

According to the Quantitative Research Design framework, this study translates the research problem and objectives into a structured analytical plan.

3.3.1 Design Type

This research relies on a Data-driven Analytics Design (Predictive Modeling Pathway). This design was selected to move from descriptive analysis to predict manufacturing failures using historical sensor data.

3.3.2 Unit of Analysis

the unit of analysis is a "Single Manufacturing Run". Each instance in the study represents a unique production cycle that allow making an investigation of sensor behavior.

3.3.3 Execution Pathway & Environment

The study organized in an analytical line that it starts with high-dimensional log extraction then feature engineering (addressing the Time Dimension and sensor correlations), and a supervised learning ensemble.

3.3.4 Rationale for Design Choice

This design has been chosen because it is the same path that used in bench mark and because of the nature of the study

3.4 Data and Measurement Procedures

This section introduces the dataset information and the practical details that followed to reach to the results

3.4.1 Dataset Description

In this study, the UCI-SECOM dataset was used, downloaded from Kaggle and originally published in the UCI Machine Learning Repository. The dataset consists of 1567 instances and 590 sensor-based features collected from the semiconductor manufacturing process. The goal of this study is to solve a binary classification problem (Pass/Fail). The researcher chooses this dataset because it is the same dataset that used in the benchmark.

3.4.2 Data Preprocessing and Cleaning

The dataset was processed to make sure that data has high quality and improve model performance. The time column was removed to focus on important sensor measurements. All sensor readings were converted to numeric format, and missing values were filled using the median to maintain the model's robustness against outliers. Columns with zero variance were removed to reduce noise and computational complexity. The imbalance between the Fail and Pass samples makes it necessary to use the SMOTE technique to increase the number of samples in the minority class. This step was essential to improving the model's ability to detect rare faults and ensure reliable predictions.

3.4.3 Model Training

The dataset was then split into 80% training and 20% testing sets using stratified sampling to preserve class distributions. the models that used are XGBoost and Random Forest and they trained carefully to capture complex patterns and ensure stable predictions. These models chosed because of their ability to handle high-dimensional sensor data and capture non-linear relationships effectively.

3.4.4 Ensemble Learning

A soft voting ensemble applied to combine the strengths of both models. This approach reduces individual model errors and improves overall classification performance..

3.4.5 Performance Evaluation

The final model evaluated using several metrics, including accuracy, precision, recall, and F1-score. Overall, these preprocessing and modeling steps ensure a robust fault detection system capable of achieving high accuracy and reliability.

3.5 Expected Outcomes

This section presents the proposed methodology that's used to predict the tolerance of the semiconductor manufacturing process. It introduced a combined models that used to classify tolerances during the stages of manufacturing to predict it earlier. It expected to reach more than 98% The macro and weighted averages of precision, recall, and F1-score expect to reach 0.99, confirming the robustness and stability of the proposed ensemble framework. These results will indicate that the model effectively detects manufacturing faults at early stages with minimal misclassification. The proposed methodology will achieve a high accuracy of 98% in predicting tolerances during the manufacturing stages. An expected excellent performance because of several reasons. First, the combined model combined together to exceed effectively the strengths of individual classifiers. Second, the preprocessing and feature selection steps ensured that the model focused on the most relevant information, minimizing noise and redundant data that could negatively affect classification. Analysis of the class-wise performance indicates that both classes (0 and 1) were classified with high precision and recall, with F1-scores of 1.00 and 0.99 respectively. This demonstrates that the methodology is capable of reliably detecting defective and non-defective items, which is critical in high-stakes manufacturing processes where early detection of anomalies can reduce cost and improve efficiency. When compared with existing studies in the field, the proposed approach shows superior accuracy and balanced performance, highlighting the effectiveness of combining multiple classification models. Previous studies often relied on single-model approaches or simpler ensemble techniques, which led to lower overall performance or difficulties in handling imbalanced datasets. Despite these strengths, some limitations should be acknowledged. The model has been evaluated on a specific dataset from the semiconductor manufacturing process, which may limit generalizability to other types of manufacturing data. Additionally, while the ensemble model provides high accuracy, it also introduces increased computational complexity and may require optimization for real-time applications.

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